Statistical Analysis of Federal Interest Rate Data

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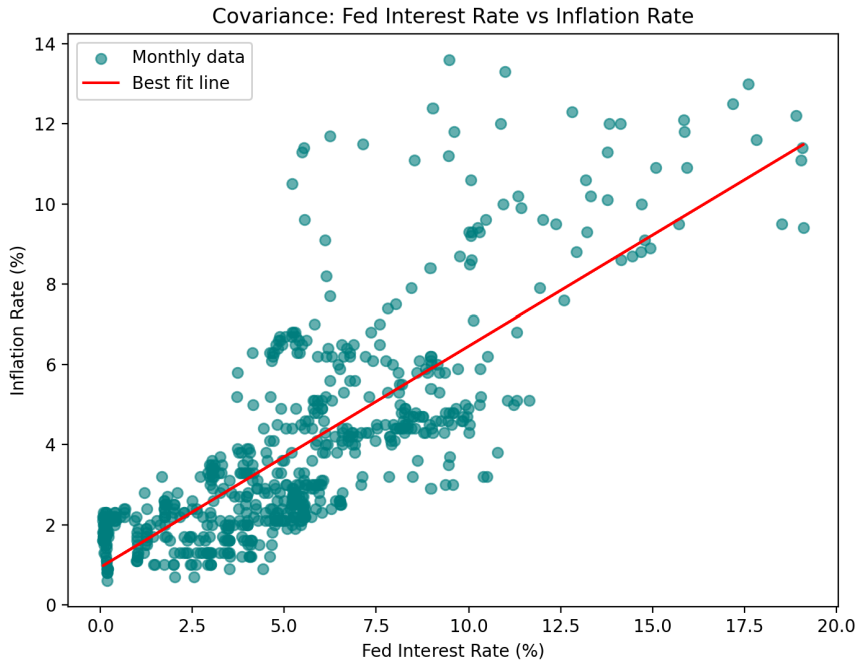
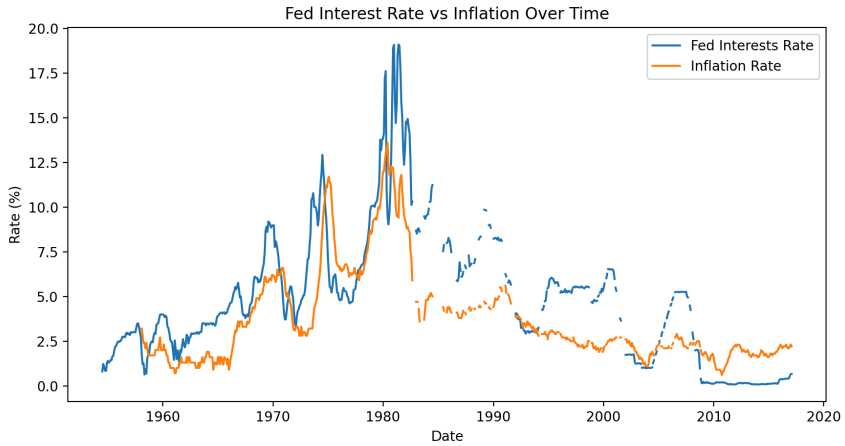
Summary:

In this project, I analyzed over 60 years of monthly Federal interest/inflation rate data using Python. I computed key statistics such as mean, variance, and standard deviation to quantify variability, and relationships within the dataset. Covariance between Federal interest rates and inflation was also calculated, and provides insight into how changes in one data set are associated with changes in the other over time. Graphical tools, including line plots, rolling averages, lines of best fit, heat maps, and scatter plots, helped make these patterns easier to interpret by highlighting trends, periods of stability, and sudden deviations from expected behavior.

Methods / Findings:

Covariance between the Federal interest rate and inflation was calculated, revealing a strong positive linear relationship. This is also reflected in the corresponding time series graph, where the two series move closely together over time.





<u>All calculations after this are based on inflation due to richer data</u>

To capture how the data behaves over time I implemented:

<u>Markov chain model</u>: Each month was assigned a state based on inflation surprises, (categorized as negative, neutral, or positive relative to recent trends).

```
state
1    555
2     81
0     74
```

The image above represents the observations per each regime on the last day of available data. This distribution of observations across regimes confirms that the neutral regime is the most frequent state. Although the 'Inflation Over Time' graph displays significant historical spikes, (most notably in the late 1970s), these represent relatively short lived periods of high volatility. When considering the full 60 year sample, the data exhibits high regime persistence, spending the vast majority of the time in a low variance and stable state.

From these regime observations a **transition matrix** is computed.

```
Transition Matrix:
          0         1         2
0  0.716216  0.270270  0.013514
1  0.036101  0.920578  0.043321
2  0.012346  0.296296  0.691358
```

(Transition matrix for last day of available data)

For example, the neutral regime held going into the next month 92 percent of the time (conditional on the current state), negative shocks held 72 percent of the time (conditional on the current state), and positive shocks 69 percent of the time (conditional on the current state). This indicates that most months experience stable rates, while extreme deviations are less common and often revert to neutral rather than immediately flipping to the opposite extreme.

<u>Regime statistics</u> were used to quantify the mean and standard deviation of surprises/shocks within each state. It was observed that negative shocks averaged μ = **–0.94**(outcomes far below expectations), neutral months clustered near **0**(outcomes were close to expectations), and positive shocks averaged μ = **+0.88**(outcomes exceeded expectations).

```
Regime Statistics:
             mean       std  count
state
0      -0.940631  0.514250     74
1      -0.008412  0.186233    554
2       0.881481  0.489753     81
```

Chebyshev analysis was used to measure the frequency of unusually large deviations within each regime. Using k = 2 the upper bound for my example was 25%, but the observed violations were much less frequent. For instance, in the negative shock regime, only about 5.4 percent of months exceeded two standard deviations from the regime mean, in the neutral regime about 2.3 percent did, and in the positive shock regime about 6.2 percent did. This confirms that extreme surprises are rare and that the regimes effectively capture the majority of variation in the data.

```
Chebyshev Violation Rates:
 state
0    0.054054
1    0.023423
2    0.061728
```

Moment generating functions were calculated for each regime to provide a complete statistical summary of the surprises/regimes. MGFs capture not only the mean and variance but also higher order moments such as skewness or kurtosis(tailedness), offering a full description of the probability distribution for each regime. These functions are useful for forecasting and risk assessment, allowing the estimation of probabilities for large deviations conditioned on the current regime.

	1	0	2
-1.0	1.025902	2.972455	0.455045
-0.9	1.021718	2.626950	0.488978
-0.8	1.017893	2.329601	0.526092
-0.7	1.014425	2.072924	0.566775
-0.6	1.011313	1.850675	0.611473
-0.5	1.008553	1.657636	0.660705
-0.4	1.006145	1.489438	0.715074
-0.3	1.004086	1.342417	0.775284
-0.2	1.002377	1.213496	0.842163
-0.1	1.001015	1.100085	0.916684
0.0	1.000000	1.000000	1.000000
0.1	0.999332	0.911395	1.093476
0.2	0.999011	0.832707	1.198742
0.3	0.999038	0.762612	1.317742
0.4	0.999412	0.699981	1.452815
0.5	1.000134	0.643855	1.606773
0.6	1.001206	0.593413	1.783016
0.7	1.002628	0.547953	1.985664
0.8	1.004403	0.506872	2.219731
0.9	1.006532	0.469650	2.491326
1.0	1.009017	0.435841	2.807927

<u>Example:</u> State 0 and state 2 have similar standard deviations, but their MGF's are mirror images. This shows that the magnitude of variability is similar, but the direction of extreme events widely differs.

Chebyshev treats them as the same, but MGFs tell us they are opposites in risk character:

- state 0: fatter tails as $t < 0$ means higher probability for extreme negative values.
- State 2: fatter tails as $t > 0$ means higher probability for extreme positive values.

Let's say you wanted to place a bet on inflation rates going up. Imagine you are currently in **state 2**. From the transition matrix, we see that once the system enters state 2, it is most likely to **remain in that same state** for the next period. This gives confidence that the positive shock environment is not fleeting.

Transition Matrix:

	0	1	2
0	0.716216	0.270270	0.013514
1	0.036101	0.920578	0.043321
2	0.012346	0.296296	0.691358

At that point, we can use state 2's MGF for insight. Its rapid growth for positive values of t indicates a higher probability of large upside surprises, meaning gains tend to come from occasional strong moves rather than small, steady increases. Chebyshev's inequality would only tell us a conservative bound on deviations of either side(25%), but the MGF shows that the risk is skewed toward the upside.

Using both together, you could size or set risk limits knowing that downside extremes are less likely than upside extremes in this regime. This supports allowing profits to run while keeping stops relatively tighter, since the dominant risk in state 2 is missing a large positive move rather than suffering an unusually large negative one.

Room for Improvement**:**

It would've been nice to be able to run these metrics on large live stock api's. With richer data, you can go from descriptive/regime analysis, to actionable, probability driven trading strategies where MGF's guide expected tail behavior, and Markov chains guide the probability of regime continuation. This approach would allow backtesting that demonstrates the model's potential to generate alpha from the selected assets.

Conclusion:

Overall, this combination of methods provides a quantitative picture of how Federal interest rates behave over time. Summary statistics and graphs reveal long term trends and short term volatility. Markov chains describe the likelihood of remaining in or transitioning between different regimes. Chebyshev analysis identifies extreme deviations relative to typical behavior within each regime. Moment generating functions offer a complete probabilistic characterization of the data. Covariance analysis shows how Federal interest rates and inflation move together. In conjunction, these techniques turn raw historical data into actionable insights, illustrating both typical behavior and rare events in a measurable way.